

SQL Random Insert Queries

120 sample records across 6 related tables

Purpose: Ready-to-use data for joins, aggregations, window functions, sales and profitability analysis, employee hierarchy, and customer segmentation.

Execution order: departments, employees, customers, products, orders, order_items. The sequence follows the supplied DDL and foreign key dependencies.

Record count: 20 rows per table.

1. departments

```
INSERT INTO departments
(department_id, department_name, location)
VALUES
(1, 'Sales', 'New York'),
(2, 'Marketing', 'Chicago'),
(3, 'Finance', 'Dallas'),
(4, 'Human Resources', 'Atlanta'),
(5, 'IT', 'Seattle'),
(6, 'Operations', 'Phoenix'),
(7, 'Customer Support', 'Denver'),
(8, 'Research', 'Boston'),
(9, 'Procurement', 'Houston'),
(10, 'Legal', 'Washington'),
(11, 'Administration', 'Miami'),
(12, 'Engineering', 'San Jose'),
(13, 'Quality Assurance', 'Austin'),
(14, 'Business Analytics', 'Charlotte'),
(15, 'Logistics', 'Detroit'),
(16, 'Training', 'Orlando'),
(17, 'Compliance', 'Philadelphia'),
(18, 'Product Management', 'San Diego'),
(19, 'Strategy', 'Nashville'),
(20, 'Security', 'Portland');
```

2. employees

```
INSERT INTO employees
(employee_id, employee_name, department_id, manager_id, salary, hire_date, job_title)
VALUES
(101, 'John Smith', 1, NULL, 120000.00, '2018-01-15', 'Sales Director'),
(102, 'Emma Johnson', 2, NULL, 115000.00, '2019-03-20', 'Marketing Director'),
(103, 'Michael Brown', 3, NULL, 118000.00, '2017-07-10', 'Finance Director'),
(104, 'Sophia Davis', 4, NULL, 110000.00, '2018-05-30', 'HR Director'),
(105, 'James Wilson', 5, NULL, 130000.00, '2016-09-14', 'IT Director'),
(106, 'Olivia Miller', 6, 101, 85000.00, '2020-02-12', 'Operations Manager'),
(107, 'William Moore', 7, 101, 75000.00, '2021-04-22', 'Support Manager'),
(108, 'Ava Taylor', 8, 105, 90000.00, '2019-06-18', 'Research Lead'),
(109, 'Benjamin Anderson', 9, 103, 80000.00, '2020-08-25', 'Procurement Manager'),
(110, 'Isabella Thomas', 10, 103, 95000.00, '2018-11-16', 'Legal Counsel'),
(111, 'Lucas Jackson', 11, 104, 65000.00, '2021-01-19', 'Admin Officer'),
(112, 'Mia White', 12, 105, 100000.00, '2019-09-21', 'Engineering Manager'),
(113, 'Ethan Harris', 13, 112, 72000.00, '2022-03-02', 'QA Lead'),
(114, 'Charlotte Martin', 14, 103, 88000.00, '2020-07-08', 'Business Analyst'),
(115, 'Alexander Thompson', 15, 106, 70000.00, '2021-05-15', 'Logistics Supervisor'),
(116, 'Amelia Garcia', 16, 104, 67000.00, '2022-06-11', 'Training Specialist'),
(117, 'Daniel Martinez', 17, 103, 89000.00, '2020-12-07', 'Compliance Officer'),
(118, 'Harper Robinson', 18, 112, 105000.00, '2019-02-14', 'Product Manager'),
(119, 'Matthew Clark', 19, 103, 97000.00, '2018-10-03', 'Strategy Manager'),
(120, 'Evelyn Rodriguez', 20, 105, 92000.00, '2021-08-09', 'Security Manager');
```

3. customers

```
INSERT INTO customers
(customer_id, customer_name, city, state, country, segment, signup_date, sales_rep_id)
VALUES
(201, 'Acme Corp', 'New York', 'NY', 'USA', 'Corporate', '2021-01-10', 101),
(202, 'BlueWave Inc', 'Chicago', 'IL', 'USA', 'Corporate', '2021-02-14', 101),
(203, 'GreenTech Ltd', 'Dallas', 'TX', 'USA', 'SMB', '2020-11-08', 106),
(204, 'Bright Retail', 'Atlanta', 'GA', 'USA', 'Consumer', '2022-04-01', 107),
(205, 'Urban Traders', 'Seattle', 'WA', 'USA', 'SMB', '2021-07-22', 101),
(206, 'Silverline LLC', 'Phoenix', 'AZ', 'USA', 'Corporate', '2020-09-13', 106),
(207, 'Prime Services', 'Denver', 'CO', 'USA', 'Corporate', '2021-08-19', 107),
(208, 'Nova Solutions', 'Boston', 'MA', 'USA', 'SMB', '2022-02-05', 101),
(209, 'Speed Logistics', 'Houston', 'TX', 'USA', 'Corporate', '2020-12-11', 106),
(210, 'Elite Stores', 'Miami', 'FL', 'USA', 'Consumer', '2021-03-25', 107),
(211, 'Future Vision', 'San Jose', 'CA', 'USA', 'Corporate', '2022-05-07', 101),
(212, 'Golden Supplies', 'Austin', 'TX', 'USA', 'SMB', '2021-06-17', 106),
(213, 'Metro Market', 'Detroit', 'MI', 'USA', 'Consumer', '2022-01-15', 107),
(214, 'Vertex Systems', 'Orlando', 'FL', 'USA', 'Corporate', '2020-10-28', 101),
(215, 'Cloud Networks', 'Philadelphia', 'PA', 'USA', 'Corporate', '2021-09-16', 106),
(216, 'Peak Ventures', 'San Diego', 'CA', 'USA', 'SMB', '2022-03-09', 107),
(217, 'Summit Retail', 'Nashville', 'TN', 'USA', 'Consumer', '2021-04-20', 101),
(218, 'Pioneer Group', 'Portland', 'OR', 'USA', 'Corporate', '2020-08-07', 106),
(219, 'Rapid Trade', 'Charlotte', 'NC', 'USA', 'SMB', '2022-06-02', 107),
(220, 'Infinity Corp', 'Washington', 'DC', 'USA', 'Corporate', '2021-11-11', 101);
```

4. products

```
INSERT INTO products
(product_id, product_name, category, subcategory, price, cost, supplier_id)
VALUES
(301, 'Laptop Pro 15', 'Electronics', 'Laptops', 1500.00, 1100.00, 1),
(302, 'Laptop Air 13', 'Electronics', 'Laptops', 1200.00, 850.00, 2),
(303, 'Wireless Mouse', 'Electronics', 'Accessories', 30.00, 12.00, 3),
(304, 'Mechanical Keyboard', 'Electronics', 'Accessories', 80.00, 35.00, 2),
(305, '27 Inch Monitor', 'Electronics', 'Displays', 250.00, 160.00, 4),
(306, 'Office Chair', 'Furniture', 'Seating', 220.00, 140.00, 5),
(307, 'Standing Desk', 'Furniture', 'Desks', 450.00, 300.00, 5),
(308, 'Conference Table', 'Furniture', 'Tables', 900.00, 650.00, 6),
(309, 'Printer LaserJet', 'Electronics', 'Printers', 320.00, 220.00, 4),
(310, 'Network Router', 'Electronics', 'Networking', 180.00, 95.00, 3),
(311, 'External SSD 1TB', 'Electronics', 'Storage', 140.00, 80.00, 2),
(312, 'Webcam HD', 'Electronics', 'Accessories', 70.00, 30.00, 1),
(313, 'Projector X200', 'Electronics', 'Projectors', 600.00, 410.00, 4),
(314, 'Notebook Pack', 'Office Supplies', 'Stationery', 15.00, 5.00, 7),
(315, 'Pen Set', 'Office Supplies', 'Stationery', 10.00, 3.00, 7),
(316, 'Filing Cabinet', 'Furniture', 'Storage', 300.00, 180.00, 5),
(317, 'Desk Lamp', 'Furniture', 'Lighting', 45.00, 18.00, 6),
(318, 'Headset Pro', 'Electronics', 'Audio', 95.00, 45.00, 2),
(319, 'Smartphone X', 'Electronics', 'Phones', 850.00, 600.00, 1),
(320, 'Tablet Plus', 'Electronics', 'Tablets', 650.00, 450.00, 3);
```

5. orders

```
INSERT INTO orders
(order_id, customer_id, employee_id, order_date, status, shipping_date)
VALUES
(401,201,101,'2023-01-05','Delivered','2023-01-08'),
(402,202,101,'2023-01-12','Delivered','2023-01-15'),
(403,203,106,'2023-01-20','Delivered','2023-01-24'),
(404,204,107,'2023-02-01','Delivered','2023-02-05'),
(405,205,101,'2023-02-10','Shipped','2023-02-12'),
(406,206,106,'2023-02-18','Delivered','2023-02-22'),
(407,207,107,'2023-03-01','Delivered','2023-03-04'),
(408,208,101,'2023-03-08','Cancelled',NULL),
(409,209,106,'2023-03-15','Delivered','2023-03-18'),
(410,210,107,'2023-03-20','Delivered','2023-03-24'),
(411,211,101,'2023-04-02','Processing',NULL),
(412,212,106,'2023-04-09','Delivered','2023-04-12'),
(413,213,107,'2023-04-15','Delivered','2023-04-19'),
(414,214,101,'2023-05-01','Shipped','2023-05-04'),
(415,215,106,'2023-05-07','Delivered','2023-05-11'),
(416,216,107,'2023-05-16','Delivered','2023-05-20'),
(417,217,101,'2023-06-03','Delivered','2023-06-06'),
(418,218,106,'2023-06-10','Processing',NULL),
(419,219,107,'2023-06-18','Delivered','2023-06-22'),
(420,220,101,'2023-06-25','Delivered','2023-06-29');
```

6. order_items

```
INSERT INTO order_items
(order_item_id, order_id, product_id, quantity, unit_price, discount)
VALUES
(501, 401, 301, 2, 1500.00, 5.00),
(502, 402, 303, 10, 30.00, 0.00),
(503, 403, 305, 3, 250.00, 2.00),
(504, 404, 306, 5, 220.00, 10.00),
(505, 405, 310, 4, 180.00, 5.00),
(506, 406, 307, 2, 450.00, 8.00),
(507, 407, 302, 1, 1200.00, 3.00),
(508, 408, 314, 20, 15.00, 0.00),
(509, 409, 309, 2, 320.00, 4.00),
(510, 410, 312, 6, 70.00, 5.00),
(511, 411, 319, 3, 850.00, 7.00),
(512, 412, 318, 8, 95.00, 5.00),
(513, 413, 311, 5, 140.00, 2.00),
(514, 414, 308, 1, 900.00, 10.00),
(515, 415, 320, 2, 650.00, 6.00),
(516, 416, 304, 12, 80.00, 3.00),
(517, 417, 313, 2, 600.00, 5.00),
(518, 418, 315, 25, 10.00, 0.00),
(519, 419, 316, 3, 300.00, 8.00),
(520, 420, 317, 10, 45.00, 4.00);
```

Notes: Foreign key references are valid. NULL values are included where appropriate. Discount values represent percentages.